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LAB08: Tiling on PULP (part 2)

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Objective of the Class

Intro: Tiling

Tasks:

- Double Buffering
- Overlapping tiles with 3x3 convolutions

Deadline:

Dec 12th 2025

Programming Language: C

Lab duration: 3h

Assignment:

- Time for delivery: 2 weeks

<u>The class is meant to be interactive: coding together, on your own, and do not be afraid to ask questions!</u>

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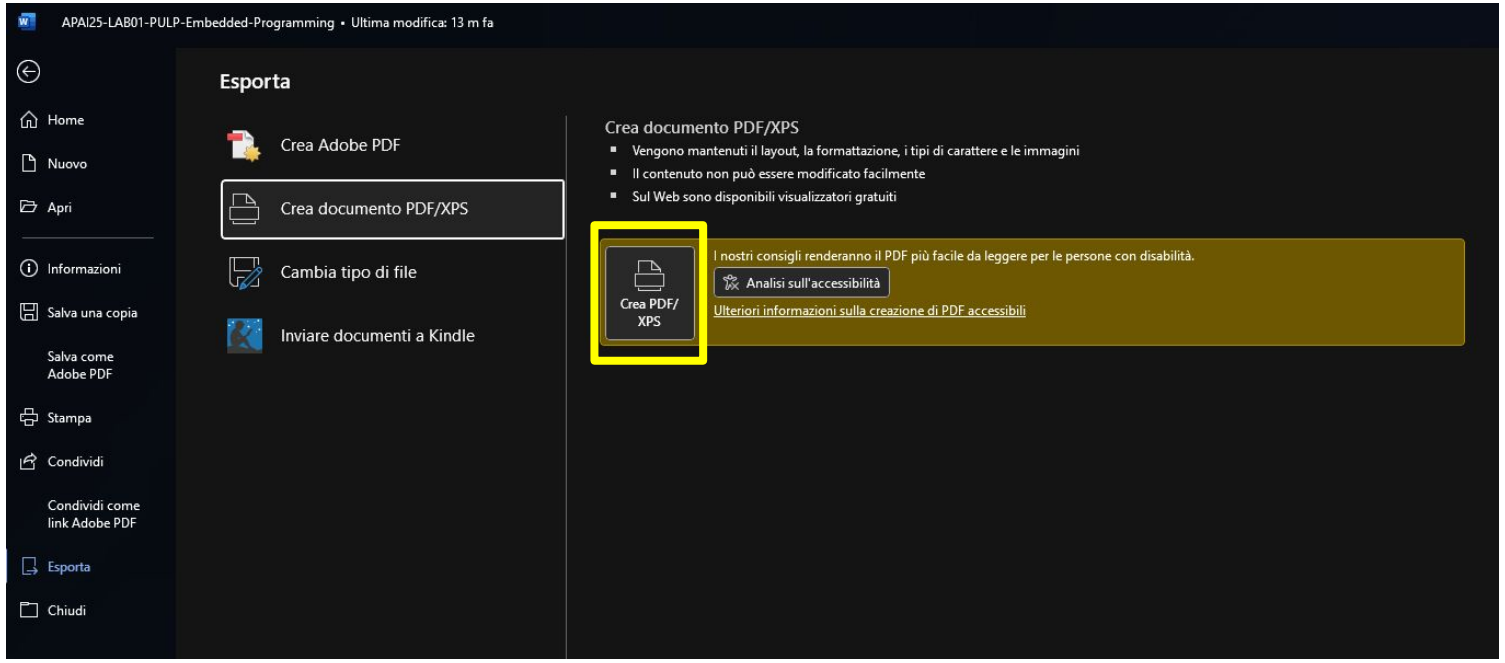
How to deliver the Assignment

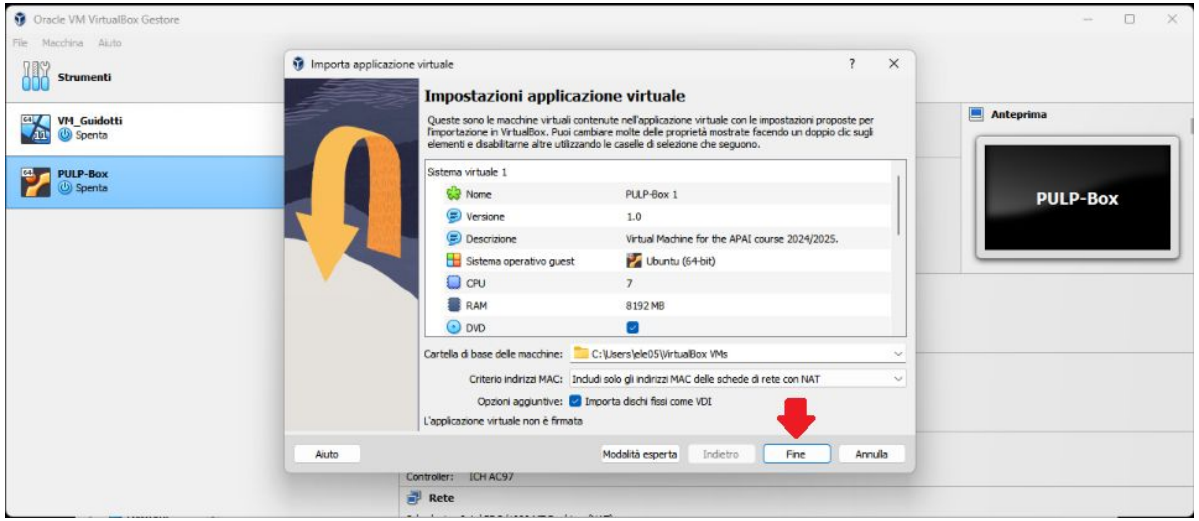
You will deliver ONLY the PDF assignment, no code

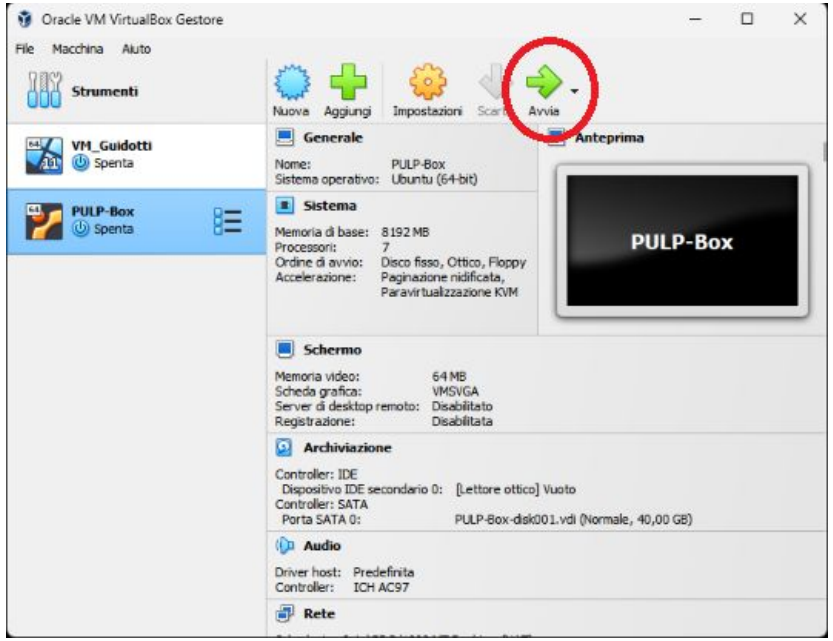
- Download the assignment file from Virtuale.
- Fill the results required by the assignment.
- Export to pdf format.
- Rename the file to: LAB<number_of_the_lesson>_APAI_<your_name>.pdf
- Use Virtuale platform to load ONLY your .pdf file



Opening the VM and VSCode

1. On the lab's PCs, open the file explorer and go to This PC, C:/VM_APAI/
2. Double click on PULP-box.ova
3. VirtualBox opens, just click on "Fine"
4. Wait for the VM to be imported
5. Open the VM with "Avvia"





Password is 'pulp'

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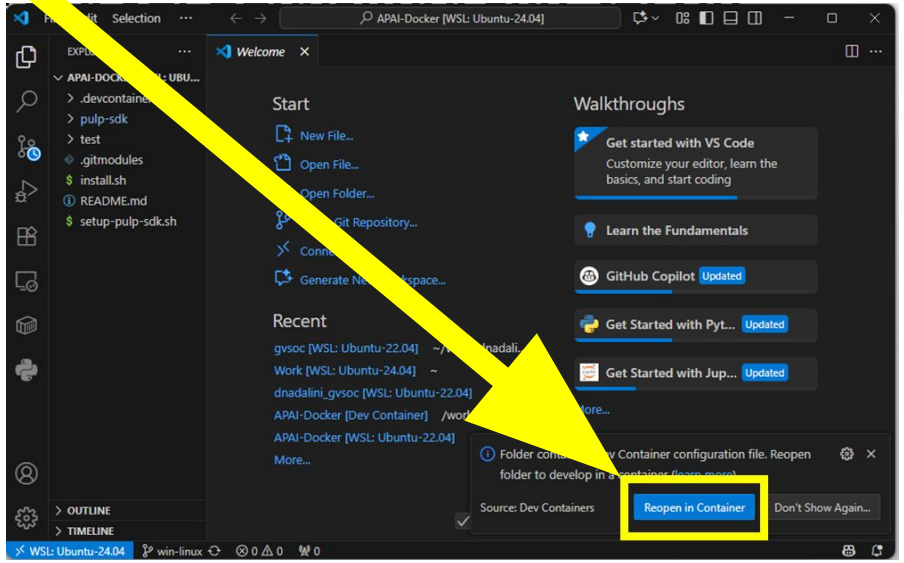
Opening the Docker with VSCode

1. Open a terminal (right click – open a new terminal)
2. From the terminal, open VSCode in the folder of the Docker

```
$ cd APAI-Docker
$ code .
```

3. Reopen the APAI-Docker folder in VSCode (click on “Reopen in container”)
4. Now you can use the integrated terminal (open with CTRL+J) to run your applications!



IMPORTANT: every time you open a new terminal to work on PULP, launch

```
$ source setup-pulp-sdk.sh
```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Getting Started:

IMPORTANT: activate the pulp-sdk module file <u>every</u> time a new shell is open.

```
$ source setup-pulp-sdk.sh
```

For the lab of today, we need to:

```
$ pip install mako numpy six
```

HOW TO RUN THE CODE:

```
$ git clone https://github.com/EEESlab/APAI25-LAB08-PULP-Tiling-Part2
$ cd APAI25-LAB08-PULP-Tiling-Part2
$ source initialization.sh
$ python3 parameters_generate.py --kernel-shape=<add_here>
--channels=<add_here> --output-spatial-dimension=<add_here>
$ make clean all run
```


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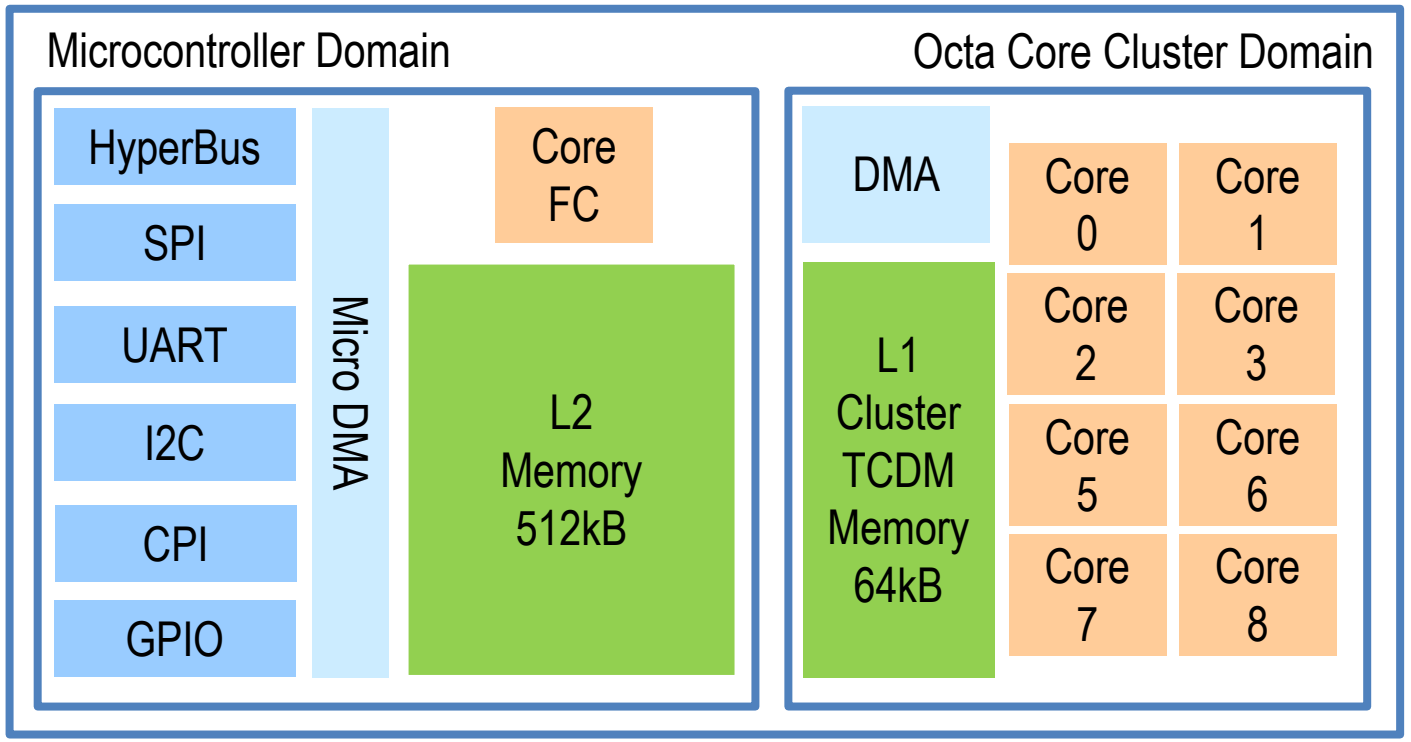
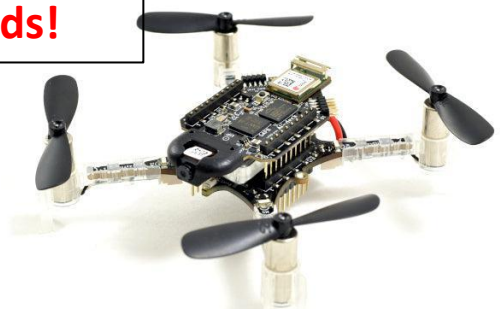
TASK4: double buffering

PULP Platform: today we focus on the 8-cores cluster



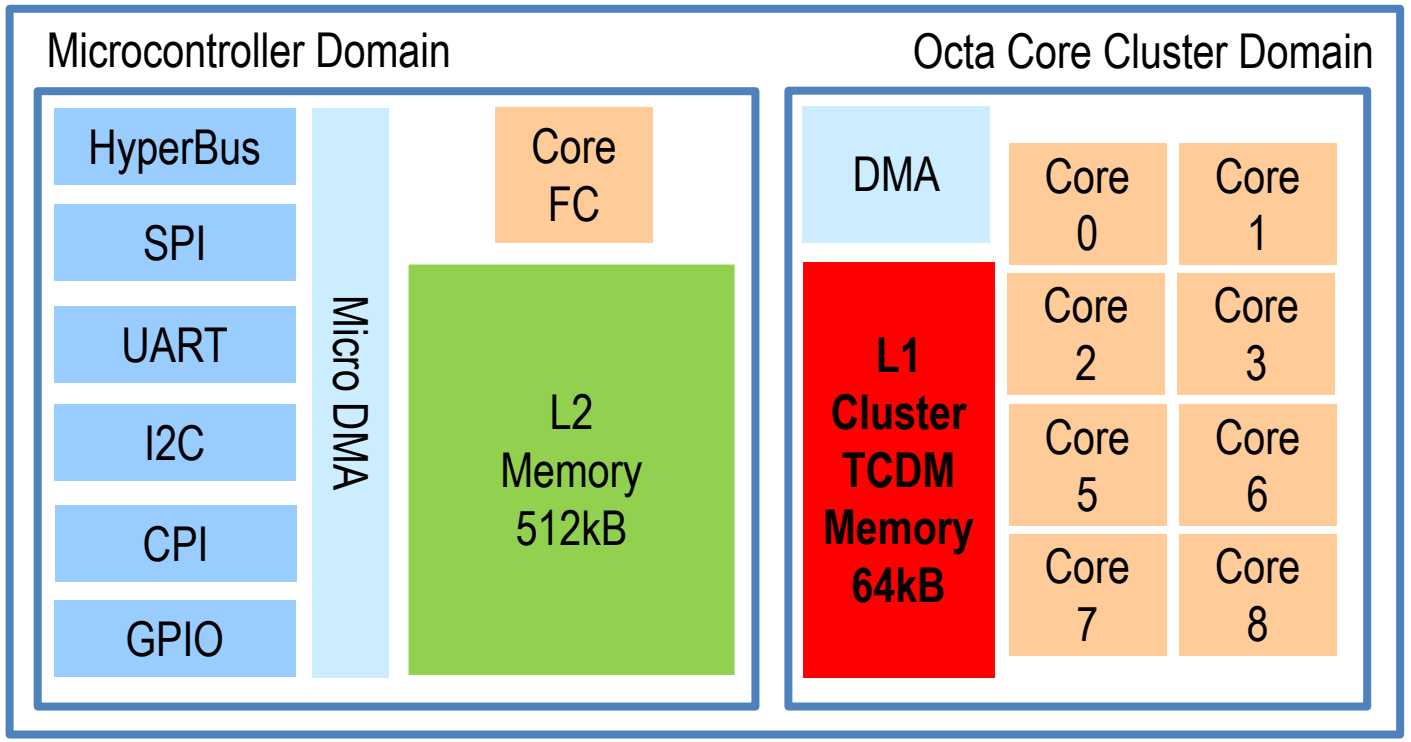
- **Cores**: 1 + 8
- **On-chip Memories**
  - A level 2 Memory, shared among all cores
  - A level 1 Memory, shared by the 8-cores cluster
- **cluster-DMA:** A multi-channel 1D/2D DMA, controlling the transactions between the L2 and L1 memories
- **micro-DMA**: A smart, lightweight and completely autonomous DMA () capable of handling complex I/O scheme
  - **Bus+Peripherals:** HyperBus, I2S, CPI, timers, SPI, GPIOs, etc…

NB: this is the architecture you find on our nano-drones and GAP boards!



GitHub HW Project: https://github.com/pulp-platform/pulp
HW Documentation: https://raw.githubusercontent.com/pulp-platform/pulp/master/doc/datasheet.pdf

PULP Platform: today we focus on the 8-cores cluster



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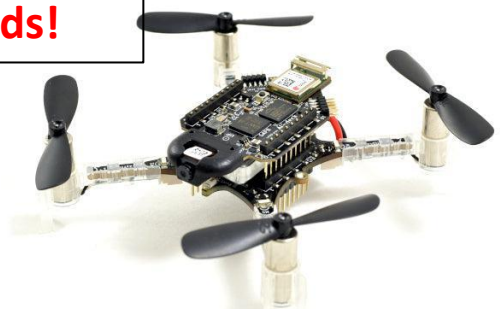

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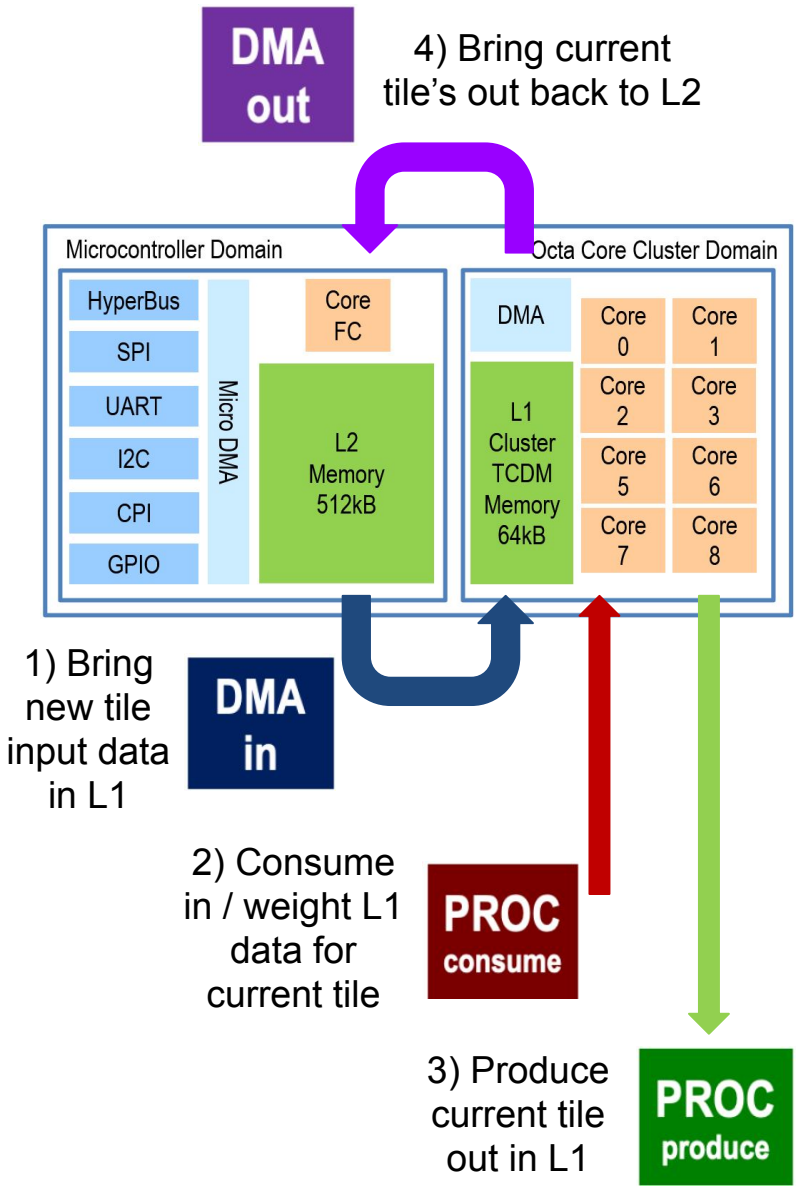
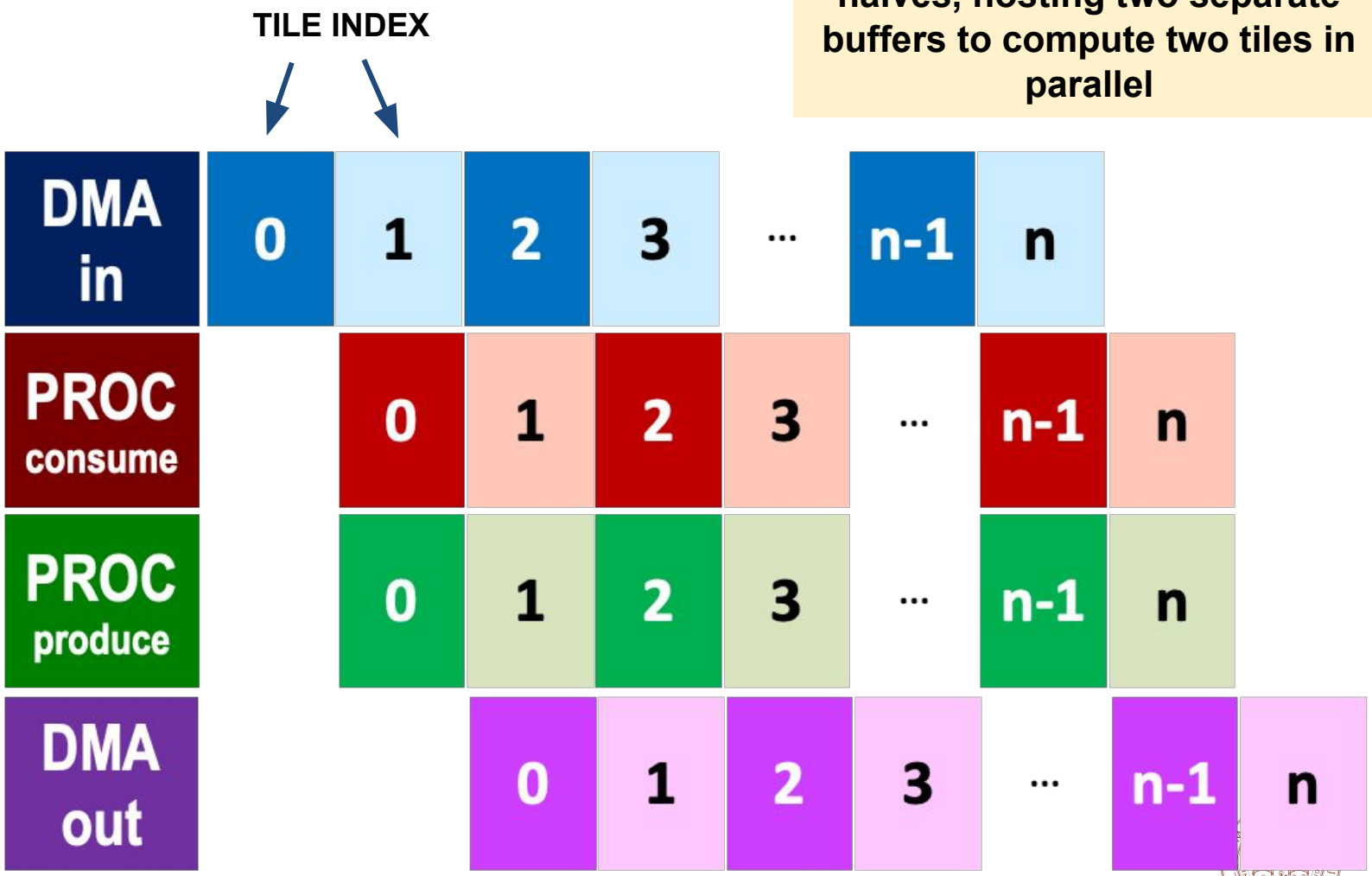
LAB07: Tiling from L2 to L1 with double buffering



DOUBLE BUFFERING: L1 memory is divided in two halves, hosting two separate buffers to compute two tiles in parallel



EX4: Double Buffering with 1x1 Conv2D

Prerequisites:

```
source setup-pulp-sdk.sh
```

Run the code:

```
1.  python3 parameters_generate.py --kernel-shape=1 --channels=<add_here> --output-spatial-dimension=<add_here>
2.  make clean all run
```

Follow the assignment document.

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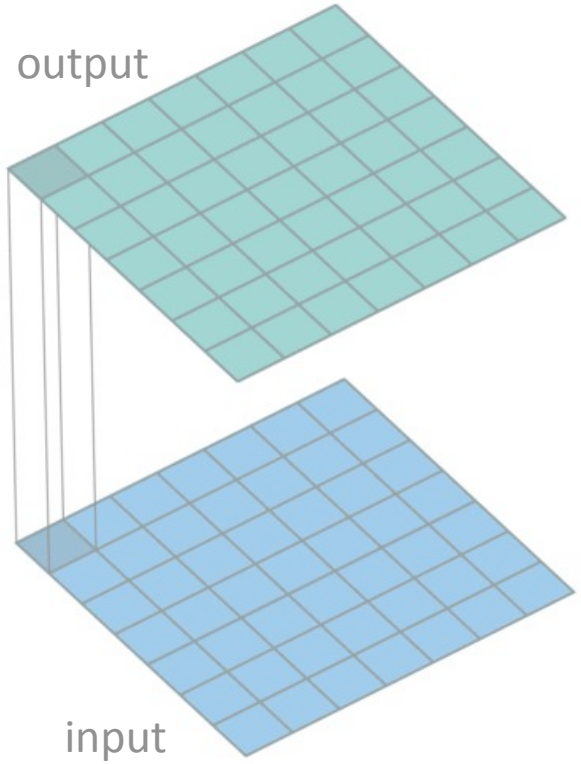
TASK5: conv3x3 and overlapping tiles

Case study: 3x3 conv2D

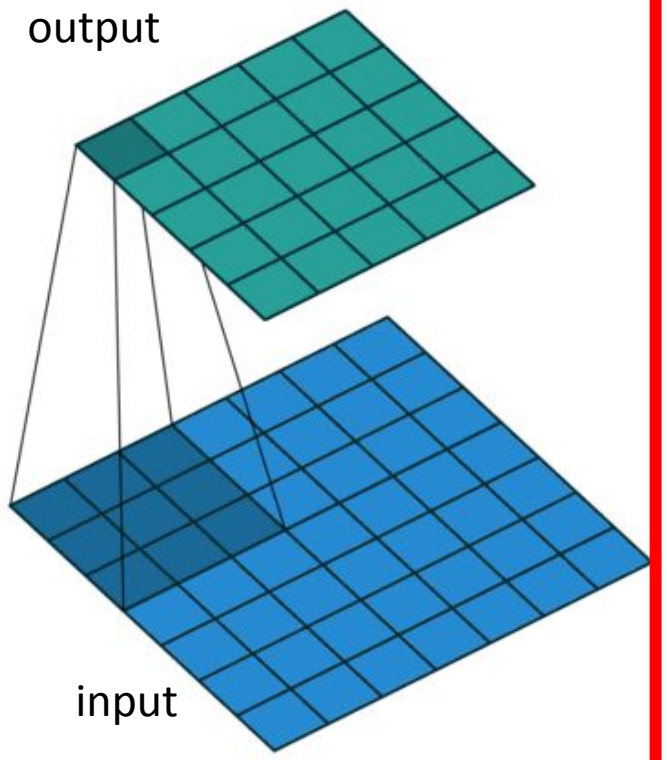
Task5.1: cov3x3

- conv1x1 the spatial size between input and output does not change!
- With conv3x3 it changes. Find out how!



1x1 Convolution

lab06



3x3 Convolution

Used in lab07!

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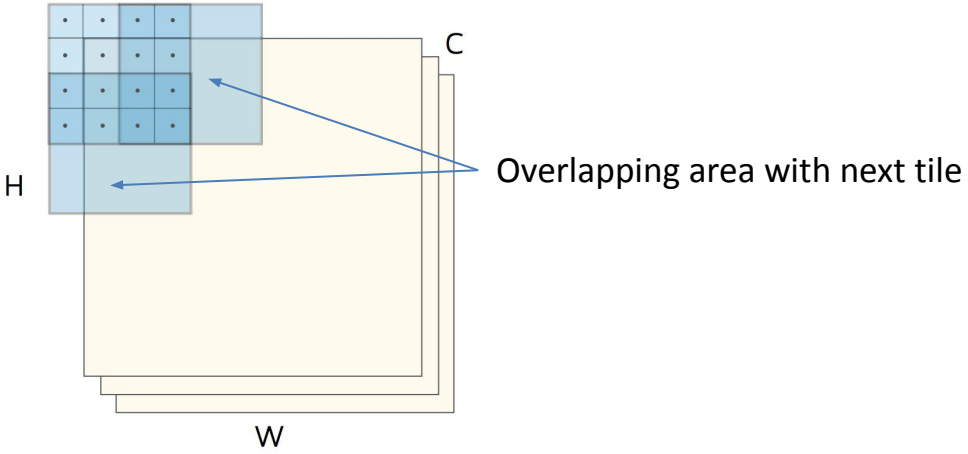
Case study: 3x3 conv2D

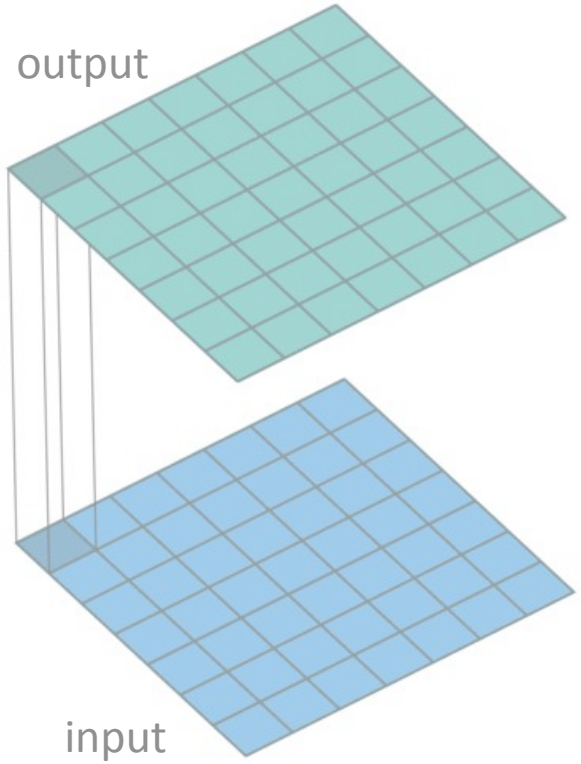
Task5.1: cov3x3

- conv1x1 the spatial size between input and output does not change!
- With conv3x3 it changes. Find out how!

Task5.2: overlapping tiles

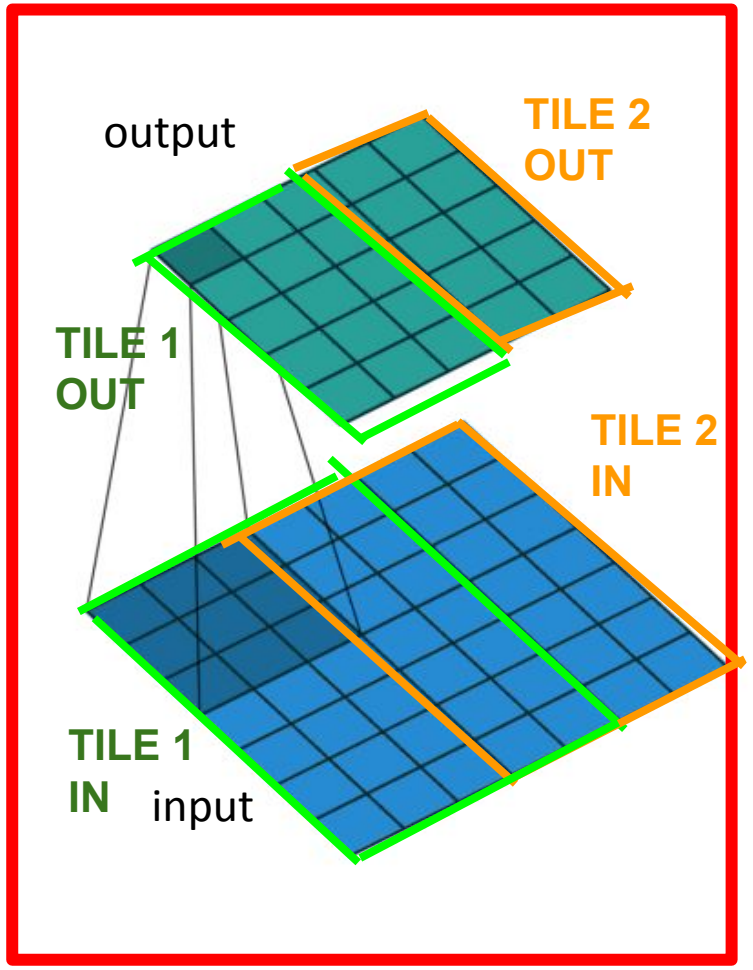
With 3x3 convolutions adjacent tiles overlap a bit, so we load the same piece of input twice.





1x1 Convolution

lab06



3x3 Convolution

Used in lab07!

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EX5: Double Buffering with 3x3 Conv2D

Prerequisites:

```
source setup-pulp-sdk.sh
```

Run the code:

```
1.  python3 parameters_generate.py --kernel-shape=3 --channels=<add_here> --output-spatial-dimension=<add_here>
2.  make clean all run
```

Follow the assignment document.

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